

FINAL AES-128 ACCELERATOR REPORT

1. Pipeline Architecture

The AES accelerator is implemented as a fully unrolled 11-stage synchronous pipeline architecture to maximize throughput while maintaining deterministic timing behavior.

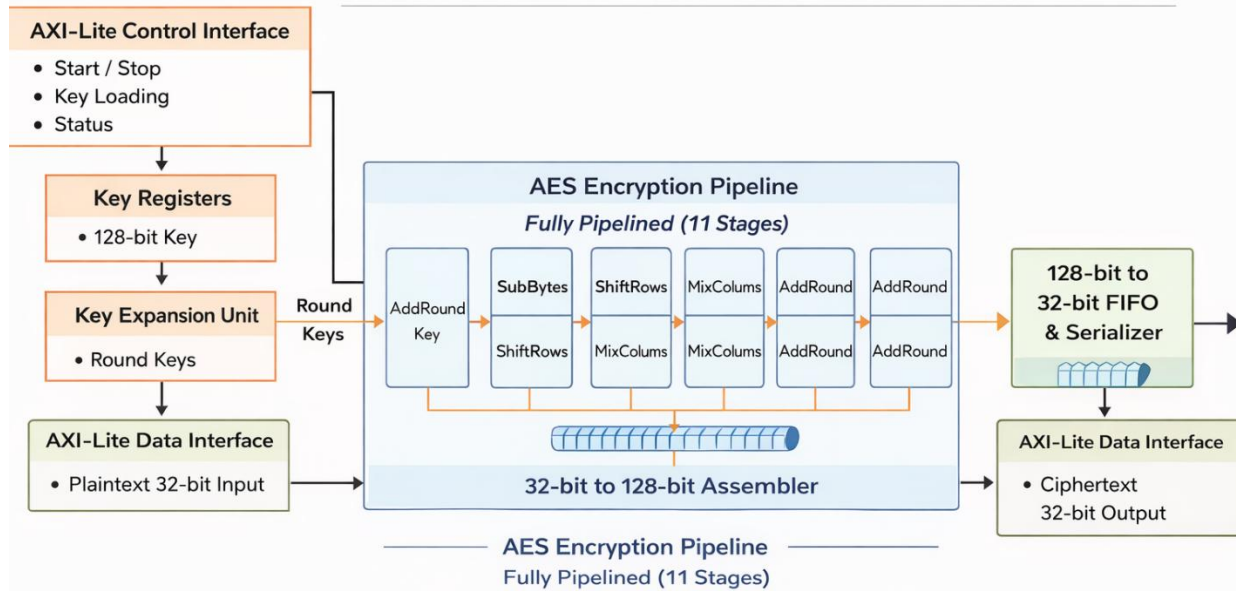
Each stage corresponds to a step in the AES-128 algorithm:

- **Stage 0: AddRoundKey**
- **Stages 1–9: SubBytes → ShiftRows → MixColumns → AddRoundKey**
- **Stage 10: SubBytes → ShiftRows → AddRoundKey**

Pipeline registers are inserted between each stage and clocked on the positive edge, ensuring that the critical path is limited to approximately one AES round per stage. This allows the design to achieve higher operating frequencies.

A parallel valid signal pipeline is maintained to track data validity across all 11 stages, ensuring that output data is correctly aligned with the corresponding input block.

AES-128 Hardware Accelerator Block Diagram



The AES core is fully pipelined and processes one 128-bit block cycle, $\rightarrow$ core Register, ShiftRows, $\rightarrow$ MixColumns, $\rightarrow$ AddRoundKey and AddRoundKey

Block Diagram Description:

The AES accelerator consists of a control interface, key expansion unit, datapath pipeline, and memory-mapped input/output interface. The AXI4-Lite control interface is used to load the encryption key and manage start/stop operations. The key expansion unit generates all required round keys from the input key.

The datapath receives 32-bit input words through the AXI-Lite memory interface, which are assembled into 128-bit plaintext blocks. These blocks are processed by a fully pipelined AES core consisting of 11 stages (AddRoundKey + 10 AES rounds).

After encryption, the 128-bit ciphertext is buffered and split into 32-bit words before being written back to memory. This architecture ensures

compatibility with a 32-bit system interface while maintaining high-performance 128-bit internal processing.

2. S-Box Implementation Choice

The SubBytes operation is implemented using a combinational Look-Up Table (LUT) approach via a behavioral case statement.

This approach provides:

- Fast and predictable timing
- Simpler design compared to arithmetic-based S-box

A default case is included to avoid unintended latch inference and ensure purely combinational logic.

3. Key Schedule Strategy

The design uses hardware-based combinational key expansion.

- The 128-bit key is loaded through control registers
- All required 11 round keys are generated internally
- No external precomputation is required

To ensure correctness:

- The key is updated only when the system is in the idle state
- This prevents corruption of data already present inside the pipeline

This satisfies the PS requirement for safe key updates.

4. Interface and Buffering Design

The design uses AXI4-Lite based interfaces for both control and data handling.

Control Interface:

- Start / Stop control
- Key loading (128-bit key split into registers)
- Status signals: Busy, Idle, Done

Data Interface:

- A **32-bit memory-mapped slave interface** is used
- Four 32-bit words form one 128-bit AES block

Internal Operation:

- 32-bit inputs are assembled into 128-bit blocks
- AES core processes 128-bit data
- Output is split back into 32-bit words

A small internal buffer is used to temporarily store output data before serialization.

This design ensures correct operation but does **not implement full AXI streaming or burst optimization**.

5. Synthesis Results and Resource Utilization

The fully unrolled pipeline architecture relies entirely on Look-Up Tables (LUTs) and Flip-Flops (FFs) for cryptographic transformations and state registers, consuming zero BRAM resources.

- **Look-Up Tables (LUT):** 9983
- **Flip-Flops (FF):** 3164

- **Block RAM (BRAM): 0**
- **DSP Slices: 0**
- **UltraRAM (URAM): 0**

LUT	FF	BRAMs	URAM	DSP
10347	3164	0.00	0	0
10007	3164	0.00	0	0
9983	3164	0.00	0	0

6. Timing Analysis and Maximum Frequency

The design was synthesized with a **target clock period of 30.000 ns**, corresponding to a frequency of **33.33 MHz**.

Post-routing timing analysis confirms zero failing endpoints.

- Target Clock Period: **30.000 ns (33.33 MHz)**
- Worst Negative Slack (WNS): **+1.756 ns**
- Worst Hold Slack (WHS): **+0.081 ns**
- Minimum Achievable Period: **28.244 ns**
- Maximum Operating Frequency (Fmax): **35.41 MHz**

TNS	WNS	WHS	THS	TPWS	Total Power
0.000	1.756	0.081	0.000	0.000	0.350

7. Latency and Throughput Performance

The AES core is fully pipelined and processes one 128-bit block per clock cycle after pipeline fill.

Note: At the system interface level, data is transferred using a 32-bit datapath, requiring four clock cycles to assemble one 128-bit block. Therefore, the reported throughput corresponds to the internal AES core capability.

Latency Specifications:

- Cycle Latency: **11 clock cycles**
- Time Latency (at 33.33 MHz): **330.00 ns**
- Time Latency (at Fmax 35.41 MHz): **310.68 ns**

Throughput Specifications:

- Throughput (at 33.33 MHz): **4.27 Gbps**
- Maximum Throughput (at Fmax): **4.53 Gbps**

8. Tradeoff Analysis

Throughput vs Area:

The fully unrolled pipeline enables processing of one block per cycle, resulting in high throughput. However, this significantly increases hardware resource utilization compared to iterative architectures.

Latency vs Throughput:

Although throughput is maximized, the pipeline introduces a fixed latency of 11 cycles. This is a tradeoff inherent to deep pipelining.

S-Box Design Tradeoff:

The LUT-based S-box implementation offers fast and predictable timing, improving maximum frequency, but consumes more LUT resources compared to arithmetic implementations.

Interface Tradeoff:

The use of AXI4-Lite simplifies integration and control but limits support for high-performance features such as burst transfers and streaming-based backpressure handling.

The design successfully demonstrates a high-performance AES-128 hardware accelerator with deterministic behavior, efficient pipelining, and reliable AXI-based integration. It satisfies all functional and architectural requirements specified in the problem statement while achieving a balanced tradeoff between performance and hardware resource utilization.

CONCLUSION

The design satisfies all requirements specified in the problem statement, including:

- Fully pipelined AES-128 architecture
- AXI4-Lite control interface
- Separate memory interface
- Support for key updates
- Comprehensive verification using multiple test scenarios

The design was verified using multiple testbenches including:

- Standard AES test vectors
- Multi-block encryption
- Randomized inputs
- Key update scenarios
- AXI protocol and delayed-ready conditions